

ENGINEERING ETHICS

Building a Responsible Future





Do Now

**In your digital notes for today
respond to the following questions**

- What does the word ethics mean to you?
- Give an example of something that fits your definition.

Day 1: Engineering Ethics



What is Responsibility?

Passive Responsibility

- Definition: Being accountable for outcomes, specifically undesirable ones, after they have occurred.
- Focus: Blame, liability, and justifying actions already taken.
- Behavior: Reactive, waiting for instructions, or avoiding responsibility.
- Example: An engineer facing legal liability after a bridge has failed.



Active Responsibility

- Definition: Taking initiative and accountability for tasks, duties, or potential outcomes before they occur.
- Focus: Prevention, care, and “making things right” in the future.
- Behavior: Proactive, forward-thinking, and action-oriented (e.g., looking for things that need to be done in a home).
- Example: An engineer ensuring a bridge is safe before it is built.

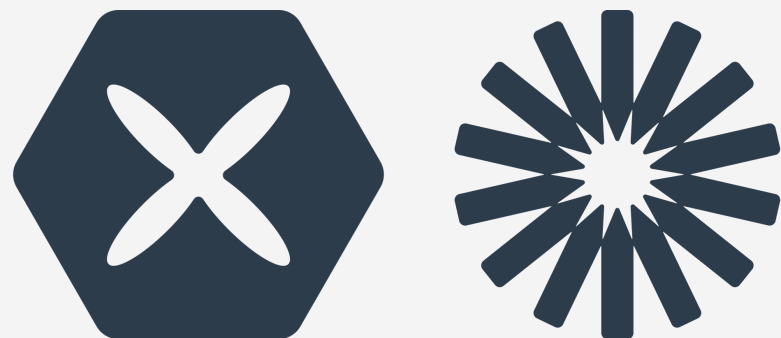


Conditions of Blameworthiness

Blameworthiness is a specific form of passive or backward-looking responsibility that becomes relevant after an undesirable event has occurred. It means that it is considered proper to blame an individual or an institution for their actions or the resulting consequences. The degree to which someone is blamed is determined by the extent to which the following four conditions are fulfilled:

- Wrong-doing
- Causal Contribution
- Foreseeability
- Freedom of Action

If an individual is unable to provide a satisfactory account of their actions regarding these conditions, they are considered accountable and target for blame.



Conditions of Blameworthiness

Wrong-doing:

This condition applies when a person has violated a norm.

These norms may be legal rules, standards common within an organization, or moral principles found in ethical frameworks like codes of conduct or normative theories.

What is an example of wrong doing based on this definition?



Conditions of Blameworthiness

Causal Contribution:

- The individual must have made a causal contribution to the undesirable consequences. It is important to note that a failure to act (an omission) is considered a causal contribution just as much as an active decision. While one person's contribution is rarely the sole cause of a disaster, it is usually a necessary ingredient in the chain of events; this means that without that person's specific action or failure to act, the consequence would not have occurred.

What is an example of casual contribution based on this definition?



Conditions of Blameworthiness

Foreseeability:

A person can only be held responsible if they were able to know the consequences of their actions. Blame is not assigned if it is "totally unreasonable" to expect someone to have been aware of the risks. However, the law and ethics expect individuals to do everything reasonably possible to become acquainted with potential consequences. In the Challenger case, this condition was fulfilled because the engineers and managers were aware of the erosion risks of O-rings at low temperatures, even if they lacked "conclusive scientific evidence" of a total failure.

What is an example of foreseeability based on this definition?



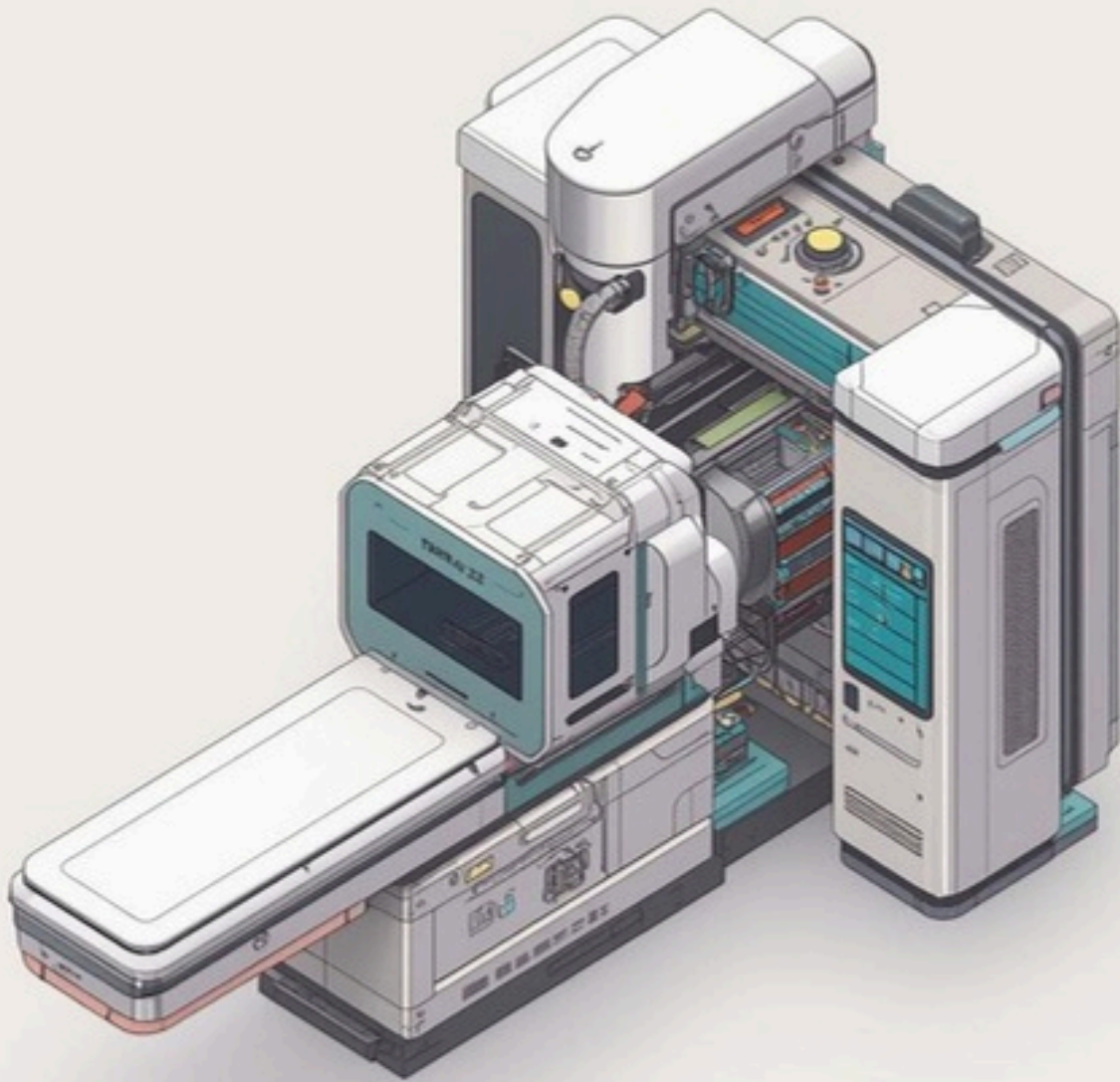
Conditions of Blameworthiness

Freedom of Action

The person must have had the freedom to act and must not have acted under compulsion or coercion. While intense pressure—such as the threat of being fired—may diminish an individual's responsibility, they are still considered responsible if they technically had the ability to act differently.

What is an example of freedom of action based on this definition?





Activity—Ethical 'Gut Check' Survey (The Challenger Case)

- Survey Question: Should an engineer's primary loyalty be to their employer or to the safety of the public?.
- Instruction: Pick a stance and write 2-3 sentences justifying your stance. Do not discuss your answer with anyone until prompted.

The Challenger Disaster

Situation

- Listen to the audio discussing what happened during the Challenger disaster. Answer the questions on the handout as you listen.



Prompt:

Which moment was the biggest ethical failure?

Ignoring O-ring warnings

Launch pressure from schedule / PR

Managers overriding engineers

Normalization of deviance

Be ready to explain why.



Group Roles (Engineering Team Model)

Each group assigns:

- **Safety Engineer**
 - Focus: Risk, public safety, harm prevention
- **Project Manager**
 - Focus: Deadlines, stakeholders, resources
- **Ethics Officer**
 - Focus: Moral responsibility, long-term consequences
- **Evidence Specialist**
 - Focus: Transcript evidence
 - NGSS Lens





Discussion Round 1

Defining the Real Failure

Question:

- What was the main failure?
 - Engineering design
 - Decision-making system
 - Organizational culture

NGSS Lens:

- What constraints existed?
- Which constraints were prioritized incorrectly?

Discussion Round 2

Responsibility in Complex Systems

Question:

Who held the most ethical responsibility?

- Engineers
- Managers
- Organization
- External pressure (government, public, media)

Must Include:

- Risk trade-off
- Data limitations
- Safety vs schedule decision





Discussion Round 3

Redesign the Decision System

If you redesigned the decision process, what would you change?

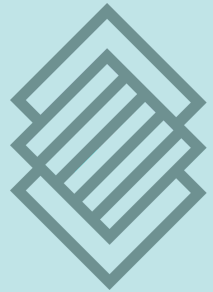
Choose at least one:

- Safety protocol
- Reporting structure
- Data requirements
- Whistleblower protection

NGSS Connection:

How does your solution reduce risk while still meeting mission goals?

Whole Class Discussion



**When does
following
procedure become
unethical?**



**Can “doing your
job” still cause
harm?**



**Should engineers
be responsible for
decisions they
don’t control?**



**How should risk be
handled when data
is incomplete?**

Final Reflections (ET)

- What is the difference between being blamed after a disaster and preventing one?
- Which of the four blameworthiness conditions were clearly met in this case?
- Can an engineer be morally responsible even if they lack final authority?

